Type MS · State the symptoms and effects of inherited conditions

Answer 1 of 4 · Question 6 part b.ii · 4 marks

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

6	(a)		An observable characteristic/trait of an individual		1
	(b)	(i)	any two from • caused by a mutation; • (giving a) recessive allele; • both/two alleles need to be present/homozygous recessive genotype;		2
		(ii)	any four from • produces/build up of (thick sticky) mucus (on membranes); • in lungs/small intestine/pancreas/sperm ducts; • coughing/difficulty breathing/less gas exchange/oxygen absorbed; • reduced (aerobic) respiration/less energy; • reduced digestion/ absorption of nutrients; • infertility in males;		4
		(iii)	• some people are carriers/have a heterozygous genotype; • one allele passed on from each parent;		2
				Total for Question 6 = 9 marks	

Type MS · State the symptoms and effects of inherited conditions

Answer 2 of 4 · Question 3 part a.iii · 1 mark

2024 JUN · Paper 01 · find part a.iii in this mark scheme extract

3 (a)	(i)	nucleus;		(1)
	(ii)	allele;		(1)
	(iii)	too many fingers/toes;		(1)
(b)	(i)	heterozygous/heterozygote/carrier;		(1)
	(ii)	Y;		(1)
(c)		Any four from: • double helix/two strands; • consists of nucleotides; • four bases; • each strand held together by hydrogen bonds/bases linked by hydrogen bonds; • complementary base pairing/A-T, G-C;		(4)

Total Question 3 = 9 marks

Type MS · State the symptoms and effects of inherited conditions

Answer 3 of 4 · Question 9 part b.ii · 1 mark

2024 JUN · Paper 01 · find part b.ii in this mark scheme extract

9 (a)	(i)	A, B, AB;	2 marks for all 3 correct, 1 mark for 1 or 2 correct	(2)
	(ii)	any five from • blood type O; • no antigens present; • (person) contains anti-A and anti-B antibodies; • other blood groups have A/B antigens; • immune response if mixed; • causes agglutination of red blood cells;		(5)
(b)	(i)	• (faulty allele) carried/found on female sex/X chromosome; • males affected/female carriers/occurs more in males;		(2)
	(ii)	• blood fails to clot/excessive bleeding;		(1)

Total for Question 9 = 10 marks

Type MS · State the symptoms and effects of inherited conditions

Answer 4 of 4 · Question 4 part c.ii · 1 mark

2024 JUN · Paper 02 (regional R) · find part c.ii in this mark scheme extract

4 (a)	(i)	11, 31, 30		(1)
	(ii)	31 (× 100); 100 = 31%;	send to review if other than 31	(2)
(b)	(i)	**D (alternative forms of the same gene in a genotype)** The following are incorrect because: A two dominant genes are the same type of allele B two recessive genes are the same type of allele C alleles are not different genes		(1)
	(ii)	• sperm and egg/gametes each contain one allele/haploid; • sperm and egg/gametes fuse; • during fertilisation; • (fertilisation) brings alleles together (in diploid cell)/chromosomes (from each gamete) are paired/one allele comes from each parent;		(4)
(c)	(i)	Hh/heterozygous; hh/homozygous recessive;	either order any suitable letters to be used for alleles	(2)
	(ii)	wheezing/coughing/shortness of breath/low weight / fatigue/over production of mucus/diarrhoea/male infertility;	allow other valid symptoms	(1)

Total for Question 4 = 11 marks